I have the following assignment instruction:
Assignment 10B: More JSON Practice
The Nobel Prize organization provides public APIs for accessing structured Nobel Prize data.
Using one or both of the APIs available at the Nobel Prize Developer Zone, your task is to use JSON data to investigate and answer four interesting, data-driven questions.

At least one of your questions should go beyond simple counts and require joining, filtering, or comparing fields across the data (e.g., “Which country lost the most Nobel laureates—born there but awarded as a citizen of another country?”).

Requirements
Use one or both Nobel Prize APIs to retrieve data in JSON format.

In R, load and transform the JSON data into tidy data frames.

Formulate four questions that can be answered from the data.

For each question:

Describe the question

Show the code used

Present the answer (table, summary, or plot)

Deliverables
Submit a single Quarto (.qmd) file that includes:

Your four questions

All R code used to retrieve and process the data

The resulting answers

Help me setup with this in R

Alright—this is a very clean assignment if you set it up right from the start. I'll give you a **ready-to-use Quarto + R workflow** so you can focus on analysis instead of setup headaches.

🔧 1. Choose the Nobel Prize API

Use the official Nobel Prize API:

Base endpoints (JSON):

- Laureates:

  `https://api.nobelprize.org/2.1/laureates`
- Prizes:

  `https://api.nobelprize.org/2.1/nobelPrizes`

You'll **likely use BOTH** (important for joins → better questions).

📦 2. Install & Load Required Packages

```r
install.packages(c("tidyverse", "httr2", "jsonlite"))

library(tidyverse)
```

‹/› R

🌐 3. Retrieve JSON Data

```r
# Laureates data
laureates_url <- "https://api.nobelprize.org/2.1/laureates"

laureates_json <- request(laureates_url) |>
  req_perform() |>
  resp_body_json()

# Prizes data
```

```r
prizes_url <- "https://api.nobelprize.org/2.1/nobelPrizes"
```

R

🧹 4. Convert JSON → Tidy Data Frames

Laureates

R

Expand nested fields:

R

Prizes

R

Expand laureates inside prizes:

‹/› R

🔗 5. Join Data (IMPORTANT for higher-level question)

```
combined_df <- prizes_df %>%
```

‹/› R

This step is **key** for your "advanced" question requirement.

🧠 6. Example Questions You Can Use

Here are **4 strong, assignment-ready questions**:

✅ Q1 (Basic)

Which countries have produced the most Nobel laureates?

‹/› R

```
  count(birth.place.country.en, sort = TRUE)
```

✅ Q2 (Time Trend)

How has the number of Nobel Prizes awarded changed over time?

```
prizes_df %>%
  count(awardYear) %>%
  ggplot(aes(x = as.numeric(awardYear), y = n)) +
```

‹/› R

✅ Q3 (Category Analysis)

Which Nobel Prize category has the most awards?

‹/› R

✅ Q4 (Advanced – REQUIRED)

Which countries have the most "brain drain"?
(Born in one country but affiliated with another)

```r
combined_df %>%
  filter(!is.na(birth.place.country.en),
         !is.na(affiliations)) %>%
```

R

👉 This satisfies:

- filtering ✅
- comparison ✅
- joining ✅

📜 7. Quarto Template (Copy This)

```markdown
---
title: "Assignment 10B: Nobel Prize JSON Analysis"
format: html
---

## Introduction
This analysis uses the Nobel Prize API to explore trends and patterns in Nobel la

## Data Collection

```{r}
```

Markdown
```

Data Cleaning

```{r}
```

Question 1

Which countries have produced the most Nobel laureates?

```{r}
```

Answer:
(Explain result)

Question 2

How has the number of Nobel Prizes changed over time?

```{r}
```

Answer:

Question 3

Which category has the most Nobel Prizes?

```{r}
```

Answer:

Question 4

Which countries experience the most “brain drain”?

```{r}
```

Answer:

```
---

# ⚠️ Common Pitfalls (avoid these)

- Nested JSON not unnested → leads to errors
- Forgetting `unnest_wider()` vs `unnest_longer()`
- Country fields sometimes missing → use `filter(!is.na())`
- Column names like `category.en` → must use exact names

---

# 🚀 If You Want Next Step
```

```
I can:
- :contentReference[oaicite:1]{index=1}
- :contentReference[oaicite:2]{index=2}
- :contentReference[oaicite:3]{index=3}
- Or :contentReference[oaicite:4]{index=4}

Just tell me 👍
```

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AI for Data Analysis Guide
Learn how to integrate AI tools for faster insights.

Got the following error:
Error in `as_tibble()`:
! Columns 1, 2, 3, 4, 5, and 20 more must be named.
Use `.name_repair` to specify repair.
Caused by error in `repaired_names()`:
! Names can't be empty.
✖ Empty names found at locations 1, 2, 3, 4, 5, etc.
Run `rlang::last_trace()` to see where the error occurred.

Yeah—this is a **very common issue with the Nobel API JSON**. The structure isn't a clean rectangular table, so `as_tibble()` chokes because some fields don't have proper names.

Let's fix it properly instead of patching it blindly.

✅ 🔍 Why This Error Happens

The JSON you pulled looks something like:

```
‹/› R
```

But inside it:

- Some elements are **nested lists**
- Some fields **don't have consistent names**
- Some entries are **NULL or unnamed**

👉 So `as_tibble()` doesn't know how to assign column names.

✅ ✅ FIX (Reliable Method)

Instead of directly using `as_tibble()`, use `jsonlite::fromJSON()` with flattening.

🔧 Step 1 — Replace Your API Code

```r
library(jsonlite)

laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
```

‹/› R

✅ Why This Works

- `flatten = TRUE` automatically:
  - expands nested lists
  - assigns proper column names like:
    - `birth.place.country.en`
    - `category.en`
- avoids the unnamed column issue entirely

🧹 Step 2 — Convert to Tibble (Optional)

```r
library(tidyverse)
```

R

Now it should work **without errors**.

🔍 Step 3 — Inspect Structure (IMPORTANT)

Before analysis, always check:

R

This helps you see:

- correct column names
- nested leftovers (if any)

⚠️ If You STILL Want to Use Your Original Method

You *can* patch it like this:

R

But honestly:

👉 This is a workaround, not a clean solution

👉 You'll still struggle later with nested data

💡 Pro Tip (Will Save You Time)

For this assignment:

- Use `fromJSON(..., flatten=TRUE)` ✅
- Then use `tidyverse` for analysis ✅

Avoid mixing:

- `httr2` + manual parsing ❌ (more fragile for this API)

🚀 Next Step

If you want, send me your `glimpse()` output and I'll:

- Identify the **exact columns to use**
- Help you write **clean joins for your advanced question**
- Or build **ready-to-submit answers**

This dataset is messy—but once cleaned, it's actually perfect for your assignment.

give me one piece working code- import library to showing head of both dataset

Here's a **single clean, working R script** from start to finish:

```r
# Install packages (run once if not installed)
install.packages(c("jsonlite", "tibble"))

# Load libraries
library(jsonlite)
library(tibble)

# Load Nobel Prize data (JSON → flattened data frames)
laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
  flatten = TRUE
)$nobelPrizes

# Convert to tibble (optional but cleaner)
laureates_df <- as_tibble(laureates_df)
prizes_df <- as_tibble(prizes_df)

# Show first few rows
```

R

This will:

- load data correctly (no naming errors)
- flatten nested JSON
- display the first rows of both datasets

If this runs without errors, you're fully set up 👍

Here is print of first dataframe:

```
# A tibble: 25 × 70
   id    fileName  gender sameAs links nobelPrizes
knownName.en knownName.se givenName.en
givenName.se familyName.en familyName.se fullName.en
fullName.se birth.date birth.year
   <chr> <chr>     <chr>  <list> <lis> <list>      <chr>
<chr>        <chr>        <chr>        <chr>        <chr>
<chr>       <chr>       <chr>      <chr>
 1 745   spence    male   <chr>  <df>  <df>        A. Michael
... A. Michael ... A. Michael   A. Michael   Spence
Spence        A. Michael... A. Michael... 1943-00-00 1943
 2 102   bohr      male   <chr>  <df>  <df>        Aage N.
Bohr Aage N. Bohr Aage N.      Aage N.      Bohr          Bohr
Aage Niels... Aage Niels... 1922-06-19 1922
 3 779   ciechano... male   <chr>  <df>  <df>        Aaron
Ciech... Aaron Ciech... Aaron        Aaron        Ciechanover
Ciechanover   Aaron Ciec... Aaron Ciec... 1947-10-01 1947
 4 259   klug      male   <chr>  <df>  <df>        Aaron Klug
Aaron Klug   Aaron        Aaron        Klug          Klug
Aaron Klug  Aaron Klug  1926-08-11 1926
 5 1004  gurnah    male   <chr>  <df>  <df>
Abdulrazak ... Abdulrazak ... Abdulrazak   Abdulrazak
Gurnah        Gurnah        Abdulrazak... Abdulrazak... 1948-
00-00 1948
 6 114   salam     male   <chr>  <df>  <df>        Abdus
```

```
Salam  Abdus Salam  Abdus       Abdus       Salam
Salam        Abdus Salam Abdus Salam 1926-01-29 1926
 7 982   banerjee  male   <chr>  <df>  <df>       Abhijit
Ban... Abhijit Ban... Abhijit      Abhijit      Banerjee
Banerjee      Abhijit Ba... Abhijit Ba... 1961-02-21 1961
 8 981   abiy      male   <chr>  <df>  <df>       Abiy Ahmed
... Abiy Ahmed ... Abiy        Abiy        Ahmed Ali     Ahmed
Ali     Abiy Ahmed... Abiy Ahmed... 1976-08-15 1976
 9 843   yonath    female <chr>  <df>  <df>       Ada E.
Yona... Ada E. Yona... Ada E.       Ada E.       Yonath
Yonath        Ada E. Yon... Ada E. Yon... 1939-06-22 1939
10 866   riess     male   <chr>  <df>  <df>       Adam G.
Rie... Adam G. Rie... Adam G.      Adam G.      Riess
Riess        Adam G. Ri... Adam G. Ri... 1969-12-16 1969
# ℹ 15 more rows
# ℹ 54 more variables: birth.place.city.en <chr>,
birth.place.city.no <chr>, birth.place.city.se <chr>,
birth.place.country.en <chr>, birth.place.country.no <chr>,
#   birth.place.country.se <chr>, birth.place.cityNow.en
<chr>, birth.place.cityNow.no <chr>,
birth.place.cityNow.se <chr>, birth.place.cityNow.sameAs
<list>,
#   birth.place.cityNow.latitude <chr>,
birth.place.cityNow.longitude <chr>,
birth.place.countryNow.en <chr>,
birth.place.countryNow.no <chr>,
birth.place.countryNow.se <chr>,
#   birth.place.countryNow.sameAs <list>,
birth.place.countryNow.latitude <chr>,
birth.place.countryNow.longitude <chr>,
```

```
birth.place.continent.en <chr>,
#   birth.place.continent.no <chr>, birth.place.continent.se
<chr>, birth.place.locationString.en <chr>,
birth.place.locationString.no <chr>,
#   birth.place.locationString.se <chr>, wikipedia.slug
<chr>, wikipedia.english <chr>, wikidata.id <chr>,
wikidata.url <chr>, death.date <chr>, death.place.city.en
<chr>, ...
# ℹ Use print(n = ...) to see more rows
```

Good—that means your data loaded **correctly** 👍
Now let's make it **usable for your assignment**, because right now it's still messy (nested columns like `nobelPrizes`, `links`, etc.).

✅ What You Have (Important Columns)

From your output, these are the most useful columns:

- `id` → unique identifier (VERY important for joins)
- `knownName.en` / `fullName.en` → name
- `gender`
- `birth.year`
- `birth.place.country.en` → country (you'll use this a LOT)
- `nobelPrizes` → ⚠️ **nested (needs unnesting)**

🔧 Step 1 — Clean & Select Useful Columns

Run this:

```r
library(tidyverse)

laureates_clean <- laureates_df %>%
  select(
    id,
    name = knownName.en,
    gender,
    birth_year = birth.year,
    country = birth.place.country.en,
    nobelPrizes
  )
```

‹/› R

🩹 Step 2 — Unnest Nobel Prize Info (VERY IMPORTANT)

Right now `nobelPrizes` is a list column. You need to expand it:

```r
laureates_expanded <- laureates_clean %>%
  unnest(nobelPrizes)
```

‹/› R

🔍 Step 3 — See What's Inside Nobel Prize Data

```R
```

You'll now see fields like:

- `awardYear`
- `category.en`
- `affiliations` (sometimes nested again)

🔥 Step 4 — Your First Working Analysis (Test)

Example: Laureates by Country

```R
```

⚠️ Important Insight

You actually have **two options now**:

Option A (Easier) ✅

Use ONLY `laureates_df` (since it already contains prize info)

Option B (Better for grading) ⭐

Join with `prizes_df` for deeper analysis

👉 For your assignment, I recommend:

- Use **laureates for 3 questions**
- Use **join for 1 advanced question**

🚀 Next Step (I Recommend This)

Now send me your `glimpse(prizes_df)` output.

I'll:

- ↳ show you exactly how to **join both datasets**
- ↳ give you a **perfect advanced question (guaranteed A-level)**
- and ↳ clean code tailored to YOUR dataset (not generic)

You're actually very close now—just one more step.

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Fillout Forms Alternative
Need an alternative to Fillout Forms? Try Jotform. Create free online forms in seconds.

here is second dataset print:
A tibble: 25 × 12

```
  awardYear dateAwarded prizeAmount
prizeAmountAdjusted links       laureates category.en
category.no category.se categoryFullName.en
categoryFullName.no categoryFullName.se
  <chr>     <chr>            <int>              <int> <list>
<list>    <chr>       <chr>       <chr>       <chr>
<chr>              <chr>
 1 1901      1901-11-12       150782           10833458 <df [1
× 4]> <df>      Chemistry    Kjemi       Kemi        The Nobel
Prize in... Nobelprisen i kjemi Nobelpriset i kemi
 2 1901      1901-11-14       150782           10833458 <df [1
× 4]> <df>      Literature   Litteratur  Litteratur  The Nobel
Prize in... Nobelprisen i litt... Nobelpriset i litt...
 3 1901      1901-12-10       150782           10833458 <df [1
× 4]> <df>      Peace        Fred        Fred        The Nobel
Peace Pr... Nobels fredspris    Nobels fredspris
 4 1901      1901-11-12       150782           10833458 <df [1
× 4]> <df>      Physics      Fysikk      Fysik       The Nobel
Prize in... Nobelprisen i fysi... Nobelpriset i fysik
 5 1901      1901-10-30       150782           10833458 <df [1
× 4]> <df>      Physiology ... Fysiologi ... Fysiologi ... The
Nobel Prize in... Nobelprisen i fysi... Nobelpriset i fysi...
 6 1902      1902-11-11       141847           10191492 <df [1
× 4]> <df>      Chemistry    Kjemi       Kemi        The Nobel
Prize in... Nobelprisen i kjemi Nobelpriset i kemi
 7 1902      1902-11-13       141847           10191492 <df [1
× 4]> <df>      Literature   Litteratur  Litteratur  The Nobel
Prize in... Nobelprisen i litt... Nobelpriset i litt...
 8 1902      1902-12-10       141847           10191492 <df [1
× 4]> <df>      Peace        Fred        Fred        The Nobel
```

```
Peace Pr... Nobels fredspris    Nobels fredspris
 9 1902      1902-11-11       141847           10191492 <df [1
× 4]> <df>      Physics      Fysikk      Fysik      The Nobel
Prize in... Nobelprisen i fysi... Nobelpriset i fysik
10 1902      1902-10-30       141847           10191492 <df [1
× 4]> <df>      Physiology ... Fysiologi ... Fysiologi ... The
Nobel Prize in... Nobelprisen i fysi... Nobelpriset i fysi...
# ℹ 15 more rows
# ℹ Use print(n = ...) to see more rows
```

Perfect—this looks exactly right 👍
Now let's turn this into a **fully usable dataset + join (your advanced requirement)**.

🔧 Step 1 — Clean the Prizes Dataset

```r
library(tidyverse)

prizes_clean <- prizes_df %>%
  select(
    year = awardYear,
    category = category.en,
    laureates
  )
```

‹/› R

🔧 Step 2 — Unnest Laureates (CRITICAL)

Right now `laureates` is a nested list → expand it:

```r
prizes_expanded <- prizes_clean %>%
  unnest(laureates)
```

‹/› R

🔍 Step 3 — Inspect Structure

‹/› R

You should now see:

- `id` → ⭐ THIS is the key to join
- `year`
- `category`

🔗 Step 4 — JOIN with Laureates Dataset (IMPORTANT)

Now combine both datasets:

```
combined_df <- prizes_expanded %>%
```

R

🧠 What You Now Have (Very Powerful Dataset)

Each row now contains:

- `name`
- `gender`
- `birth_year`
- `country`
- `year` (award year)
- `category`

👉 This is exactly what your assignment wants.

✅ Test It (Make Sure Everything Works)

Example 1 — Top Countries

R

Example 2 — Nobel Prizes by Category

```R
```

Example 3 — Laureates Over Time

```R
combined_df %>%
  count(year) %>%
  ggplot(aes(x = as.numeric(year), y = n)) +
```

⭐ Your ADVANCED Question (Use This — It's Strong)

"Which countries have Nobel laureates who won in multiple categories?"

```R
combined_df %>%
  distinct(name, category, country) %>%
  count(country, name) %>%
```

👉 This includes:

- grouping ✅
- filtering ✅
- comparison ✅
- join usage ✅

🚀 What You Should Do Next

You're basically done with setup.

Now:

1. Pick **4 questions** (you already have good ones now)
2. For each:
   - write 1–2 sentence description
   - include code
   - show output

If you want, I can now:

- ↳ turn this into a **full Quarto submission (ready to submit)**
- or ↳ **write the answers in clean academic style** (for better grades)

Just say 👍

This will be my approach submission quarto document. I dont need to do anything right now until data import and printing. Please create a quarto file based on the following template that I used before. Try to keep it concise and sweet:

```
---
title: "Assignment Sentiment Analysis"
author: "Khandker Qaiduzzaman"
format: html
editor: visual
---

# Objective

The goal of this assignment is to replicate the sentiment analysis example from Chapter 2 of Text Mining with R: A Tidy Approach (Silge & Robinson) and extend it using a different text corpus and additional sentiment lexicons.

-------------------------------------------------------------------------------

# Approach

This analysis follows a two-part structure:

1.  Reproduction of the Chapter 2 sentiment analysis
```

example
2. Extension using a real-world news dataset collected via an external API

Step 1: Reproducing the Chapter 2 Example

This step reproduces the sentiment analysis workflow from Chapter 2 of *Text Mining with R: A Tidy Approach* (Silge & Robinson). The chapter demonstrates sentiment analysis using tidy text principles, where text is treated as individual word tokens and sentiment is computed by joining words with sentiment lexicons.

The process assumes that overall sentiment can be estimated by aggregating word-level sentiment contributions. Text is first converted into a tidy format using `unnest_tokens()`, stop words are removed using `anti_join()`, and sentiment values are assigned through `inner_join()` with sentiment lexicons.

The analysis uses three lexicons from the `tidytext` package:

- **Bing:** positive/negative classification
- **AFINN:** numeric sentiment scores (-5 to +5)
- **NRC:** emotion categories (e.g., joy, fear, anger)

These lexicons are applied to the example dataset from Jane Austen's novels, and sentiment is summarized across words and text sections using tidy data operations such as joins, grouping, and counting.

Step 2: Extension Using NewsAPI and Additional Sentiment Lexicons

To extend the analysis, I use news headlines collected through the NewsAPI service as the external text corpus. Headlines are well-suited for sentiment analysis because they are short, structured, and represent real-world, continuously updated news content, making them a strong contrast to the literary dataset used in Chapter 2.

Initially, the New York Times API was considered; however, due to rate limits and restricted access for large-scale retrieval, I switched to NewsAPI, which allows more flexible and reliable access to a larger number of headlines.

The dataset is retrieved using the NewsAPI `/v2/everything` endpoint with keyword-based queries (e.g., politics, world, business). Only the `title` field is extracted and used as the text corpus for analysis.

To extend the sentiment analysis beyond the original example, two additional lexicons are applied alongside the

Bing lexicon:

- **NRC Lexicon:** assigns words to emotional categories such as joy, fear, anger, and trust
- **AFINN Lexicon:** provides numeric sentiment scores ranging from negative to positive intensity

These additional lexicons enable a more detailed comparison of sentiment interpretation across different methods and provide both categorical and intensity-based perspectives on the same dataset.

Data Analysis Workflow

The analysis begins by reproducing the Chapter 2 sentiment analysis workflow from *Text Mining with R*, including tokenization with `tidytext`, application of the Bing lexicon, and summarization of sentiment distribution.

For the extension, news headlines are collected using the NewsAPI `/v2/everything` endpoint, and the JSON response is converted into a tidy data frame with the `title` field used as the text corpus.

The combined dataset is then tokenized using `unnest_tokens()`, and sentiment analysis is performed using the Bing, NRC, and AFINN lexicons to compute both

sentiment counts and scores.

Finally, results are compared across lexicons and against the original Chapter 2 example to evaluate how different sentiment methods and text sources affect interpretation, particularly between structured literary text and real-world news headlines.

Anticipated Challenges

Several challenges are expected:

- News headlines are very short, which may reduce sentiment word matches
- Different lexicons may produce inconsistent sentiment classifications
- API limitations may affect the amount of data that can be retrieved
- News content may contain ambiguous or neutral language, making sentiment detection more difficult

---

Example API Implementation (NewsAPI)

```
{r}
library(httr)
library(jsonlite)
library(dplyr)

# API key
api_key <- Sys.getenv("NEWS_API_KEY")

# Endpoint URL
url <- paste0(
  "https://newsapi.org/v2/everything?q=politics&
  api_key
)

# Make request
response <- GET(url)

# Convert to JSON
data <- fromJSON(content(response, "text", enco

# Extract headlines only
headlines <- data$articles$title

# Convert to dataframe
headlines_df <- data.frame(headline = headlines

# View
head(headlines_df)
```

Here's your **concise Quarto approach document**, adapted for the Nobel Prize JSON assignment and stopping at **data import + printing** as requested:

```
---
title: "Assignment 10B: Nobel Prize JSON Analysis"
author: "Khandker Qaiduzzaman"
format: html
editor: visual
---

# Objective

The goal of this assignment is to use the Nobel Prize API to retrieve JSON data,

---------------------------------------------------------------------

# Approach

This analysis focuses on working with structured JSON data from a public API and

---------------------------------------------------------------------

## Step 1: Data Collection Using Nobel Prize API

The dataset is obtained from the :contentReference[oaicite:0]{index=0}, which pro

Two endpoints are used:

-   **Laureates Endpoint:** contains personal information about Nobel laureates (
-   **Prizes Endpoint:** contains information about Nobel Prize awards (e.g., yea

The data is retrieved in JSON format and converted into R data frames.
```

```
---------------------------------------------------------------------

## Step 2: JSON Transformation and Data Preparation

The JSON data is converted into tidy data frames using the `jsonlite` package wit

Initial inspection of the datasets is performed using `head()` to understand stru

---------------------------------------------------------------------

## Data Analysis Workflow

The workflow begins with retrieving JSON data from the Nobel Prize API and conver

The structure of both datasets is then examined using basic inspection functions

Further data cleaning, transformation, and analysis (including joins and compari

---------------------------------------------------------------------

## Anticipated Challenges

Several challenges are expected:

-   JSON data contains nested structures that require flattening and unnesting
-   Some fields may have missing or inconsistent values
-   Joining datasets correctly requires careful use of unique identifiers
-   Variable names can be complex due to nested JSON formatting

---------------------------------------------------------------------

## Example API Implementation (Nobel Prize API)

```{r}
# Load libraries
library(jsonlite)
```
```

```
library(tibble)

# Retrieve Laureates data
laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

# Retrieve Prizes data
prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
  flatten = TRUE
)$nobelPrizes

# Convert to tibble
laureates_df <- as_tibble(laureates_df)
prizes_df <- as_tibble(prizes_df)

# View first few rows
head(laureates_df)
```

Markdown

```
---

If you want next, I can help you :contentReference[oaicite:1]{index=1}.
```

I dont see four questions in your quarto file. I am giving you the instrcution again. Re generate the Quarto file and

make sure it includes my approach to solve this problem and questions:
Assignment 10B: More JSON Practice
The Nobel Prize organization provides public APIs for accessing structured Nobel Prize data.
Using one or both of the APIs available at the Nobel Prize Developer Zone, your task is to use JSON data to investigate and answer four interesting, data-driven questions.

At least one of your questions should go beyond simple counts and require joining, filtering, or comparing fields across the data (e.g., "Which country lost the most Nobel laureates—born there but awarded as a citizen of another country?").

Requirements
Use one or both Nobel Prize APIs to retrieve data in JSON format.

In R, load and transform the JSON data into tidy data frames.

Formulate four questions that can be answered from the data.

For each question:

Describe the question

Show the code used

Present the answer (table, summary, or plot)

Deliverables
Submit a single Quarto (.qmd) file that includes:

Your four questions

All R code used to retrieve and process the data

The resulting answers

You may complete this assignment individually or in a small group.

Got it—you need an **approach document that already includes the four questions**, but still only goes up to **data import + preview (no full analysis yet)**.

Here's a clean, concise Quarto file aligned with your template and assignment:

```
---
title: "Assignment 10B: Nobel Prize JSON Analysis"
author: "Khandker Qaiduzzaman"
format: html
editor: visual
---
```

```
# Objective

The goal of this assignment is to use the Nobel Prize API to retrieve JSON data,

---------------------------------------------------------------------------

# Approach

This analysis focuses on retrieving structured JSON data from the Nobel Prize AP

---------------------------------------------------------------------------

## Step 1: Data Collection Using Nobel Prize API

The dataset is obtained from the :contentReference[oaicite:0]{index=0}, which pro

Two endpoints are used:

-   **Laureates Endpoint:** includes personal details such as name, gender, and b
-   **Prizes Endpoint:** includes award details such as year and category

The data is retrieved in JSON format and converted into R data frames.

---------------------------------------------------------------------------

## Step 2: JSON Transformation and Data Preparation

The JSON data is converted into tidy data frames using the `jsonlite` package wit

Initial inspection is performed using `head()` to understand:
- Variable structure
- Key identifiers (e.g., `id`)
- Fields needed for analysis (country, category, year)

Further transformation (such as unnesting and joining datasets) will be performed
```

```
---------------------------------------------------------------------

## Data Analysis Workflow

The workflow begins by retrieving and loading JSON data into R and converting it

Next, the structure of both datasets is examined to identify important variables

Finally, the prepared dataset will be used to answer four analytical questions in

---------------------------------------------------------------------

## Research Questions

The following four questions will guide the analysis:

1. **Which countries have produced the most Nobel laureates?**
   This question explores the geographic distribution of Nobel Prize winners base

2. **How has the number of Nobel Prizes awarded changed over time?**
   This examines trends in Nobel Prize awards across years.

3. **Which Nobel Prize category has the highest number of awards?**
   This compares the distribution of awards across different categories such as P

4. **Which countries have laureates who won Nobel Prizes in multiple categories?**
   This question requires joining datasets and analyzing whether individuals from

---------------------------------------------------------------------

## Anticipated Challenges

Several challenges are expected:

- JSON data contains nested structures requiring flattening and unnesting
- Some variables may contain missing or inconsistent values
- Proper joins require careful matching using unique identifiers (`id`)
```

```
- Column names may be complex due to nested JSON formatting

---------------------------------------------------------------------------

## Example API Implementation (Nobel Prize API)

```{r}
# Load libraries
library(jsonlite)
library(tibble)

# Retrieve Laureates data
laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

# Retrieve Prizes data
prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
  flatten = TRUE
)$nobelPrizes

# Convert to tibble
laureates_df <- as_tibble(laureates_df)
prizes_df <- as_tibble(prizes_df)

# Preview data
head(laureates_df)
```

Markdown

```
---

This version:
```
```

```
- ✅ Matches your previous style
- ✅ Includes **clear approach**
- ✅ Includes **4 questions (with 1 advanced)**
- ✅ Stops at **data import + preview**
- ✅ Ready to submit as an approach doc

---

If you want next, I can :contentReference[oaicite:1]{index=1}.
```

what is :contentReferenceoaicite:0?

That `:contentReferenceoaicite:0` is **not something you wrote or need to worry about**—it's essentially a **formatting artifact / glitch** that sometimes appears when content is copied from systems that support citations or rich references.

✅ What it means (simply)

- It's a **broken or leftover citation marker**
- It does **not belong in your Quarto document**
- It has **no meaning in R or Quarto**

❌ Why it appears

It usually comes from:

- Copy-paste from AI tools or editors that use internal citation tagging
- Hidden metadata that didn't translate properly

✅ What you should do

Just **delete it** wherever you see it.

👍 Example

If you see something like:

```
The data is retrieved from the API :contentReferenceoaicite:0
```

👉 Change it to:

```
The data is retrieved from the API.
```

⚠️ Important

If you leave it in:

- It may render as weird text in your HTML output
- Your professor might think it's a mistake (which it is)

If you want, you can paste your `.qmd` here and I'll quickly clean any hidden artifacts.

can you give from where the data was collected?
The dataset is obtained from the :contentReferenceoaicite:0, which provides structured data on Nobel laureates and prize awards.

You should replace that broken reference with the actual source name.

Here's the correct sentence:

Markdown

If you want a slightly more formal version (better for grading):

Markdown

Optional (if you want to be extra clear)

You can also mention endpoints:

Markdown

If you want, I can also ↳ help you add a **proper citation/reference section** at the end of your Quarto doc.

give me direct link

Here are the **direct official links** you can use in your assignment:

🌐 Developer Zone (main page)

Use this as your primary reference:

- Nobel Prize Developer Zone ↗

👉 This is the official page describing the API and its usage. NobelPrize.org

---

📊 API Endpoints (what you actually used in R)

You can also directly mention these:

- Laureates endpoint: https://api.nobelprize.org/2.1/laureates ↗
- Prizes endpoint: https://api.nobelprize.org/2.1/nobelPrizes ↗

These endpoints return JSON data about laureates and prizes. NobelPrize.org

---

📜 Optional (Documentation / Examples)

- API Examples Page ↗

✅ What to Put in Your Quarto

Use this sentence:

Markdown

If you want, I can also give you a **proper APA/MLA citation** for your assignment.

Sources

Add this link as a hyperlink to the quarto line

Here is your Quarto line with a proper clickable hyperlink:

Markdown

If you also want the API endpoints clickable:

Markdown

Out of curiosity- I was looking into ssecond dataset and I dont see any primary key to join with first table: awardYear dateAwarded prizeAmount prizeAmountAdjusted links laureates category.en category.no category.se categoryFullName.en categoryFullName.no categoryFullName.se 1901 1901-11-12 150782 10833458 nobelPrize, https://api.nobelprize.org/2/nobelPrize/che/1901, GET, application/json 160, 1, 1, list(list(rel = "laureate", href = "https://api.nobelprize.org/2/laureate/160", action = "GET", types = "application/json")), Jacobus H. van 't Hoff, Jacobus Henricus van 't Hoff, in recognition of the

extraordinary services he has rendered by the discovery of the laws of chemical dynamics and osmotic pressure in solutions, såsom ett erkännande av den utomordentliga förtjänst han inlagt genom upptäckten av lagarna för den kemiska dynamiken och för det osmotiska trycket i lösningar Chemistry Kjemi Kemi The Nobel Prize in Chemistry Nobelprisen i kjemi Nobelpriset i kemi 1901 1901-11-14 150782 10833458 nobelPrize, https://api.nobelprize.org/2/nobelPrize/lit/1901, GET, application/json 569, 1, 1, list(list(rel = "laureate", href = "https://api.nobelprize.org/2/laureate/569", action = "GET", types = "application/json")), Sully Prudhomme, Sully Prudhomme, in special recognition of his poetic composition, which gives evidence of lofty idealism, artistic perfection and a rare combination of the qualities of both heart and intellect, såsom ett erkännande av hans utmärkta, jämväl under senare år ådagalagda förtjänster som författare och särskilt av hans om hög idealitet, konstnärlig fulländning samt sällspord förening av hjärtats och snillets egenskaper vittnande diktning Literature Litteratur Litteratur The Nobel Prize in Literature Nobelprisen i litteratur Nobelpriset i litteratur

Great observation—and you're absolutely right to question this. This is actually one of the trickier parts of the Nobel Prize API.

🔑 Short Answer

There is **no direct primary key between the two tables** like a simple shared column.

Instead, you join them using a **nested relationship inside the `laureates` field of the prizes dataset**.

🧠 How the Data is Actually Designed

You have:

1. Laureates dataset

Each person has:

- `id` ← unique identifier ⭐ (this is your primary key)

Example:

```
id = 745
```

---

2. Prizes dataset

Each prize row contains a **nested list**:

```
laureates = list of laureate IDs
```

Inside your print you saw:

```
160, Jacobus H. van 't Hoff
569, Sully Prudhomme
```

So each prize row contains:

- laureate `id`
- but stored inside a nested structure, not a flat column

🔗 Correct Join Logic

You **DO NOT join prize table directly**

Instead you do:

Step 1: Unnest prizes → extract laureate IDs

Step 2: Join using `id`

⚙️ Correct R Workflow

```r
library(tidyverse)

# Step 1: expand laureates inside prizes
prizes_expanded <- prizes_df %>%
  unnest(laureates)

# Step 2: extract laureate id from nested structure
```

‹/› R

Then join:

R

⚠️ Why it feels confusing

Because:

- API is **not relational database style**
- it's **JSON hierarchical structure**
- so relationships are **embedded, not flat keys**

💡 Mental Model (Important)

Think of it like this:

Table	Key idea
laureates	one row per person (id = PK)
prizes	one row per award
prizes$laureates	list of foreign keys hidden inside JSON

🧩 Why your intuition is correct

You said:

> "I don't see any primary key to join"

✅ Correct in SQL sense
❗ But in JSON APIs, keys are often **nested instead of explicit columns**

🚀 If you want next step

I can show you:

- the **cleanest possible final join (1-line version)**
- OR a **diagram of how Nobel API data is structured**
- OR help you build your **advanced question (this structure is perfect for it)**

Seems like you did not write this challenge in the quarto section. Can you update this section with other challenges as well:
Anticipated Challenges
Several challenges are expected:

JSON data contains nested structures requiring flattening and unnesting
Some variables may contain missing or inconsistent values
Proper joins require careful matching using unique identifiers (id)
Column names may be complex due to nested JSON formatting

Good catch—that section can definitely be strengthened because your dataset has a few *very specific JSON quirks*.

Here's an improved, more accurate version you can directly replace in your Quarto file:

Anticipated Challenges

Several challenges are expected when working with the Nobel Prize JSON data:

- JSON data contains deeply nested structures (e.g., laureates inside prizes), requiring careful flattening and unnesting using functions like `flatten`, `unnest_longer()`, and `unnest_wider()`.
- The two datasets do not have a simple relational primary key structure; joins must be constructed using nested identifiers (e.g., extracting `id` from embedded laureate records within the prizes dataset).
- Some variables contain missing or inconsistent values (e.g., unknown birth places, incomplete country information, or missing affiliations), which may require filtering or handling `NA` values during analysis.
- Column names are automatically generated from nested JSON paths (e.g., `birth.place.country.en`), which can be long, inconsistent, and difficult to work with.
- List-columns (such as `links` and `laureates`) require additional processing before they can be used in standard data analysis workflows.
- Some fields contain multi-level nested lists, making it necessary to carefully inspect data structure using functions like `glimpse()` and `str()` before transformation.

If you want, I can also help you make this section ↳ **shorter (professor-friendly)** or ↳ **more formal (report-style)** depending on your class expectation.

Actually modify the whole quarto document based on these rabit hole findings:

```
---
title: "Assignment 10B Nobel Prize"
author: "Khandker Qaiduzzaman"
format: html
editor: visual
---

# Objective

The goal of this assignment is to use the Nobel Prize API to retrieve JSON data, transform it into tidy data frames in R, and explore the dataset to answer four data-driven questions.

-------------------------------------------------------------------------------

# Approach

This analysis focuses on retrieving structured JSON data from the Nobel Prize API, transforming it into tidy data, and preparing it for analysis using tidyverse principles.
```


Step 1: Data Collection Using Nobel Prize API

The dataset is obtained from the [Nobel Prize Developer Zone](https://www.nobelprize.org/organization/developer-zone-2/), which provides structured JSON data on Nobel laureates and prize awards.

Two endpoints are used:

- **Laureates Endpoint:** includes personal details such as name, gender, and birth country\
- **Prizes Endpoint:** includes award details such as year and category

The data is retrieved in JSON format and converted into R data frames.

Step 2: JSON Transformation and Data Preparation

The JSON data is converted into tidy data frames using the `jsonlite` package with `flatten = TRUE` to simplify nested structures.

Initial inspection is performed using `head()` to understand:

\- Variable structure\
- Key identifiers (e.g., id)\
- Fields needed for analysis (country, category, year)

Further transformation (such as unnesting and joining datasets) will be performed in later steps.

Data Analysis Workflow

The workflow begins by retrieving and loading JSON data into R and converting it into tidy data frames.

Next, the structure of both datasets is examined to identify important variables for analysis. The datasets will later be cleaned, unnested, and joined using common identifiers to enable deeper analysis.

Finally, the prepared dataset will be used to answer four analytical questions involving counting, grouping, filtering, and comparison.

Research Questions

The following four questions will guide the analysis:

1. **Which countries have produced the most Nobel laureates?**\
   This question explores the geographic distribution of Nobel Prize winners based on their birth country.

2. **How has the number of Nobel Prizes awarded changed over time?**\
   This examines trends in Nobel Prize awards across years.

3. **Which Nobel Prize category has the highest number of awards?**\
   This compares the distribution of awards across different categories such as Physics, Chemistry, and Literature.

4. **Which countries have laureates who won Nobel Prizes in multiple categories?**\
   This question requires joining datasets and analyzing whether individuals from certain countries have received awards across multiple categories, involving filtering and comparison.

Anticipated Challenges

Several challenges are expected:

- JSON data contains nested structures requiring flattening and unnesting\
- Some variables may contain missing or inconsistent values\
- Proper joins require careful matching using unique identifiers (`id`)\
- Column names may be complex due to nested JSON formatting

Example API Implementation (Nobel Prize API)

```{r}
# Load libraries
library(jsonlite)
library(tibble)
library(gt)

# Retrieve Laureates data
laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

# Retrieve Prizes data
```

```
prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
  flatten = TRUE
)$nobelPrizes

# Convert to tibble
laureates_df <- as_tibble(laureates_df)
prizes_df <- as_tibble(prizes_df)

# Preview data
laureates_df |>
  head(n = 2) |>
  gt()

prizes_df |>
  head(n = 2) |>
  gt()
```

Here is your **fully revised, cleaned, and more accurate Quarto document** based on the real structure of the Nobel API (including nested JSON realities, join complexity, and your "rabbit hole" findings).

I also kept it **concise, submission-ready, and consistent with your style**.

```
---
title: "Assignment 10B Nobel Prize"
```

```
author: "Khandker Qaiduzzaman"
format: html
editor: visual
---

# Objective

The goal of this assignment is to use the Nobel Prize API to retrieve JSON data,

---------------------------------------------------------------------

# Approach

This analysis focuses on retrieving structured JSON data from the Nobel Prize API

---------------------------------------------------------------------

## Step 1: Data Collection Using Nobel Prize API

The dataset is obtained from the [Nobel Prize Developer Zone](https://www.nobelpr

Two endpoints are used:

- **Laureates Endpoint:** contains personal information such as name, gender, bir
- **Prizes Endpoint:** contains award-level information such as year, category, a

The data is retrieved in JSON format and converted into R data frames for further

---------------------------------------------------------------------

## Step 2: JSON Structure and Data Preparation

The JSON data is parsed using the `jsonlite` package with `flatten = TRUE` to par

However, the datasets still contain complex nested components such as:
- List-columns (e.g., `nobelPrizes`, `laureates`, `links`)
- Nested identifiers inside sub-structures
```

```
- Hierarchical fields generated from JSON paths (e.g., `birth.place.country.en`)

Initial inspection using `head()` and `glimpse()` is used to understand:
- Variable structure
- Key identifiers (especially `id`)
- Fields required for analysis (country, year, category)

Further data wrangling such as unnesting and restructuring will be required in la

---------------------------------------------------------------------------

## Data Analysis Workflow

The workflow begins by retrieving JSON data from the Nobel Prize API and convert

Next, both datasets are explored to understand their structure and nesting behav

Since the data is not structured in a traditional relational format, joins requi

After restructuring, the datasets will be combined to enable comparative and lon

Finally, the cleaned dataset will be used to answer four research questions invo

---------------------------------------------------------------------------

## Research Questions

The following four questions will guide the analysis:

1. **Which countries have produced the most Nobel laureates?**
   This question explores the geographic distribution of Nobel Prize winners bas

2. **How has the number of Nobel Prizes awarded changed over time?**
   This examines historical trends in Nobel Prize distribution across years.

3. **Which Nobel Prize category has the highest number of awards?**
   This compares the frequency of awards across categories such as Physics, Chem
```

```
4. **Which countries have the highest level of cross-category Nobel achievements
   This requires joining laureate and prize data and analyzing whether individua

---------------------------------------------------------------------

## Anticipated Challenges

Several challenges are expected when working with this dataset:

- JSON data contains deeply nested structures requiring flattening and unnesting
- The dataset does not follow a traditional relational database structure; joins
- Some fields contain missing or inconsistent values (e.g., unknown countries or
- Column names are generated from nested JSON paths, resulting in long and comple
- List-columns such as `links` and `laureates` require additional processing befo

---------------------------------------------------------------------

## Example API Implementation (Nobel Prize API)

```{r}
# Load libraries
library(jsonlite)
library(tibble)
library(gt)

# Retrieve Laureates data
laureates_df <- fromJSON(
  "https://api.nobelprize.org/2.1/laureates",
  flatten = TRUE
)$laureates

# Retrieve Prizes data
prizes_df <- fromJSON(
  "https://api.nobelprize.org/2.1/nobelPrizes",
  flatten = TRUE
)$nobelPrizes
```
```

```
# Convert to tibble
laureates_df <- as_tibble(laureates_df)
prizes_df <- as_tibble(prizes_df)

# Preview data
laureates_df |>
  head(n = 2) |>
  gt()

prizes_df |>
```

Markdown

```
---

If you want next, I can help you:
- :contentReference[oaicite:0]{index=0}
- or :contentReference[oaicite:1]{index=1}
- or :contentReference[oaicite:2]{index=2}
```

Your research questions are too basic. Can you come up with really interesting questions:
Research Questions

The following four questions will guide the analysis:

Which countries have produced the most Nobel laureates? This question explores the geographic distribution of Nobel Prize winners based on their country of birth.

How has the number of Nobel Prizes awarded changed over time?This examines historical trends in Nobel Prize distribution across years.

Which Nobel Prize category has the highest number of awards?This compares the frequency of awards across categories such as Physics, Chemistry, and Literature.

Which countries have the highest level of cross-category Nobel achievements? (Advanced Question)This requires joining laureate and prize data and analyzing whether individuals from the same country have received awards in multiple categories.

Good call—your current questions are basically “starter pack analytics.” We can upgrade them to something that actually shows **data thinking + joins + interpretation**, which is what your instructor is looking for.

Here are **much stronger, more interesting, and higher-grade research questions** based on the real structure of the Nobel dataset:

🚀 Improved Research Questions (A-Level Quality)

1. Geographic Inequality in Nobel Recognition (Refined)

Which countries produce Nobel laureates, and how does this change when considering *birth country vs. award affiliation country*?

👉 Why this is better:

- introduces **two geographic dimensions**
- naturally leads to **brain drain discussion**
- requires **comparison logic**

---

2. Age at Time of Award and Field Differences

How does the age at the time of receiving a Nobel Prize vary across different categories (Physics, Medicine, Literature, etc.)?

👉 Why this is better:

- requires deriving a new variable (age = award year − birth year)
- introduces **distribution analysis**
- allows boxplots or summaries

---

3. Gender Representation Over Time and Across Fields

How has gender representation among Nobel laureates changed over time, and does it differ by category?

👉 Why this is better:

- time trend + grouping
- highlights **inequality dimension**

- allows visualization + proportions

---

4. Multi-Award and Cross-Category Laureates (Advanced Join Question)

Which laureates have received Nobel Prizes in multiple categories, and what patterns exist in their country of origin and fields of recognition?

👉 Why this is better:

- requires **true joining + deduplication**
- detects rare cases (multi-category winners)
- deeper insight than simple "multiple categories by country"

🧠 Why These Are Stronger

Compared to your original set, these:

✅ Require derived variables (age, time trends)
✅ Require real joins and filtering
✅ Go beyond counting (analysis of inequality, distribution, structure)
✅ Produce meaningful stories (not just tables)
✅ Match what professors expect for "data-driven questions"

✨ If You Want Even More Polish

I can also:

- ↳ align each question with the **exact R code you'll need**
- ↳ suggest **plots for each question (boxplot, line chart, heatmap)**
- or ↳ rewrite your whole Quarto "Research Questions" section in final submission style

Just tell me 👍

Speechify · Sponsored

Research to audio
Listen to papers and articles as podcasts

Give me in quarto format for these 4 questions. you dont have to be too creative about all questions, 1/2 is enough. Also, keep an eye on technical difficulty. We want interesting question not the notoriously difficult to answer question

Here's a **clean Quarto-ready "Research Questions" section** with improved quality, but still **feasible and not overly complex** (1–2 slightly advanced, rest straightforward):

```
## Research Questions

The following four questions will guide the analysis. The questions are designed

---------------------------------------------------------------------

### 1. Which countries have produced the most Nobel laureates, and how does this

This question explores geographic distribution of Nobel laureates using two persp

---------------------------------------------------------------------

### 2. How has the number of Nobel Prizes awarded changed over time across differ

This question examines historical trends in Nobel Prize distribution and compares

---------------------------------------------------------------------

### 3. What is the distribution of age at the time of receiving a Nobel Prize acr

This question investigates whether laureates in different fields tend to receive

---------------------------------------------------------------------

### 4. Which Nobel laureates have won prizes in multiple categories?
```

Markdown

👍 Why this version works well

- ✅ Only **2 moderately advanced questions (Q1, Q4)**
- ✅ Q2 and Q3 are **very standard but meaningful**

- ✅ No overly complex joins or exotic transformations
- ✅ Fits perfectly with your dataset structure
- ✅ Easy to implement in R without struggling

If you want next, I can:

- ↳ map each question → exact R code
- or ↳ design your **final plots (ggplot plan)**
- or ↳ help you ensure this gets an **A-level rubric score**

can you organize simpler question first and advanced question last

Absolutely—here's your **reordered Quarto "Research Questions" section**, with simpler questions first and the more advanced ones at the end:

```
## Research Questions

The following four questions will guide the analysis. The questions progress fror

---------------------------------------------------------------------

### 1. How has the number of Nobel Prizes awarded changed over time across differ
```

```
This question examines historical trends in Nobel Prize distribution and compares

---------------------------------------------------------------------

### 2. What is the distribution of age at the time of receiving a Nobel Prize acr

This question explores whether laureates in different fields tend to receive the

---------------------------------------------------------------------

### 3. Which countries have produced the most Nobel laureates?

This question explores the geographic distribution of Nobel Prize winners based c

---------------------------------------------------------------------

### 4. Which countries have produced Nobel laureates, and how does this differ wh
```

Markdown

If you want, I can next:

- ↳ match each question with **exact R code blocks**
- or ↳ help you decide **which 2 questions deserve plots (for grading impact)**

Organize the following question and indicate which one is advanced question:
Research Questions

The following four questions will guide the analysis. The questions are designed to balance interpretability and analytical depth while remaining feasible using the Nobel Prize API structure.

---

1. Which countries have produced the most Nobel laureates, and how does this compare between birth country and award affiliation country?

This question explores geographic distribution of Nobel laureates using two perspectives: the country of birth and the country associated with their affiliation at the time of the award. This allows for a basic comparison of national contribution versus institutional recognition.

---

2. How has the number of Nobel Prizes awarded changed over time across different categories?

This question examines historical trends in Nobel Prize distribution and compares how awards have evolved across major categories such as Physics, Chemistry, Medicine, Literature, and Peace. It helps identify whether certain fields have become more prominent over time.

---

3. What is the distribution of age at the time of receiving a Nobel Prize across different categories?

This question investigates whether laureates in different fields tend to receive the Nobel Prize at different stages of life. Age will be derived using birth year and award year, and summarized by category.

---

4. Which Nobel laureates have won prizes in multiple categories?

This question identifies individuals who have received Nobel Prizes in more than one category. It requires grouping and filtering across the joined dataset and helps highlight rare cases of cross-disciplinary recognition.

Here is your **cleaned and properly organized Research Questions section**, with the **advanced question clearly marked and placed last**:

Markdown

```
## Research Questions
```

```
The following four questions will guide the analysis. The questions are designed

---------------------------------------------------------------------

### 1. How has the number of Nobel Prizes awarded changed over time across differ

This question examines historical trends in Nobel Prize distribution and compares

---------------------------------------------------------------------

### 2. What is the distribution of age at the time of receiving a Nobel Prize acr

This question investigates whether laureates in different fields tend to receive

---------------------------------------------------------------------

### 3. Which countries have produced the most Nobel laureates, and how does this

This question explores geographic distribution of Nobel laureates using two persp

---------------------------------------------------------------------

### 4. **(Advanced Question)** Which Nobel laureates have won prizes in multiple

This question identifies individuals who have received Nobel Prizes in more than
```

If you want, I can also:

- mark which question needs a **join (important for grading)**
- or suggest which 2 questions should include **plots vs tables** for maximum points